

HIROCK - 4D time-lapse photogrammetric monitoring of paraglacial rockfall dynamics in rapidly deglaciating alpine terrain

1. Background and motivation

Rapid glacier retreat is transforming high mountain landscapes worldwide. In the European Alps, glaciers have lost a substantial proportion of their volume since the Little Ice Age, and the rate of retreat has accelerated during recent decades due to climate warming (Huss and Hock 2018; Zemp et al. 2019). **Deglaciation exposes formerly ice-covered terrain to subaerial conditions and initiates a phase of geomorphic adjustment** commonly described as paraglacial landscape evolution (Ballantyne 2002; Heckmann et al. 2016). During this transitional phase, slopes that were formerly buttressed by glacier ice experience a reduction in mechanical support and altered thermal and hydrological regimes, frequently leading to **increased rockfall activity and rock slope instability** (McColl 2012; Gruber and Haeberli 2007). As a result, rockfall activity is increasingly recognised as a **growing hazard for alpine infrastructure, tourism, and mountain communities** (Huggel et al. 2012).

Despite the increasing recognition of these processes, **quantitative observations of rockfall dynamics in recently deglaciating environments remain limited.** Existing studies are commonly based on **episodic field surveys, repeat terrestrial laser scanning campaigns, or multi-temporal airborne surveys.** While these approaches allow the detection of cumulative surface change, they often **lack the temporal resolution required to resolve the timing of individual rockfall events or to identify their environmental triggers.**

Recent advances in photogrammetry and automated monitoring systems offer new opportunities for high-frequency observation of geomorphic change. **Multi-view photogrammetry** has become an **increasingly powerful tool for reconstructing high-resolution three-dimensional surface models** and detecting surface change in geomorphic environments (Eltner et al. 2016; James et al. 2019). Time-lapse camera networks have successfully been applied to monitor dynamic Earth surface processes such as glacier calving, coastal cliff erosion, and hillslope sediment transport (Lim et al. 2005; Young et al. 2011; How et al. 2015; Eltner et al. 2017). However, **continuous multi-camera photogrammetric monitoring of rockfall activity in rapidly deglaciating paraglacial rockwalls** has so far **rarely been implemented.**

The proposed project **HIROCK addresses this gap** by establishing an **autonomous multi-camera monitoring system capable of producing high-resolution four-dimensional datasets** of rockwall activity. By combining time-lapse imaging with photogrammetric reconstruction, the monitoring system will enable the **detection and quantification of individual rockfall events at sub-daily temporal resolution.** This approach will provide **new insights into the frequency, magnitude, and environmental drivers of rockfall activity** in recently deglaciating alpine rockwalls and contribute to improved understanding of paraglacial slope dynamics and associated natural hazards.

2. Scientific objectives

The overall objective of HIROCK is to quantify the temporal dynamics of rockfall activity in a rapidly deglaciating alpine rockwall and to assess how glacier retreat influences paraglacial slope instability. This objective will be addressed through the following research questions:

- **How does rockfall frequency vary on diurnal and seasonal time scales** in a recently deglaciated rockwall environment?
- **Which environmental drivers**, such as temperature fluctuations, precipitation events, and freeze thaw cycles, **control the timing of rockfall activity?**
- **How does the spatial distribution of rockfall relate to the distance from the glacier surface** and the progressive exposure of rockwalls during glacier retreat?
- To what extent can **time-lapse photogrammetric monitoring complement or replace traditional methods** such as terrestrial laser scanning and UAV photogrammetry in high alpine terrain?

3. Study area

The study focuses on the **Schwarze Wand rockwall above the current terminus of Gepatschferner in the Kaunertal valley in Tyrol, Austria** (46.873723° N, 10.770135° E). Gepatschferner is the second largest glacier in Austria and has experienced strong retreat over recent decades, making it one of the most rapidly changing glacier systems in the Eastern Alps.

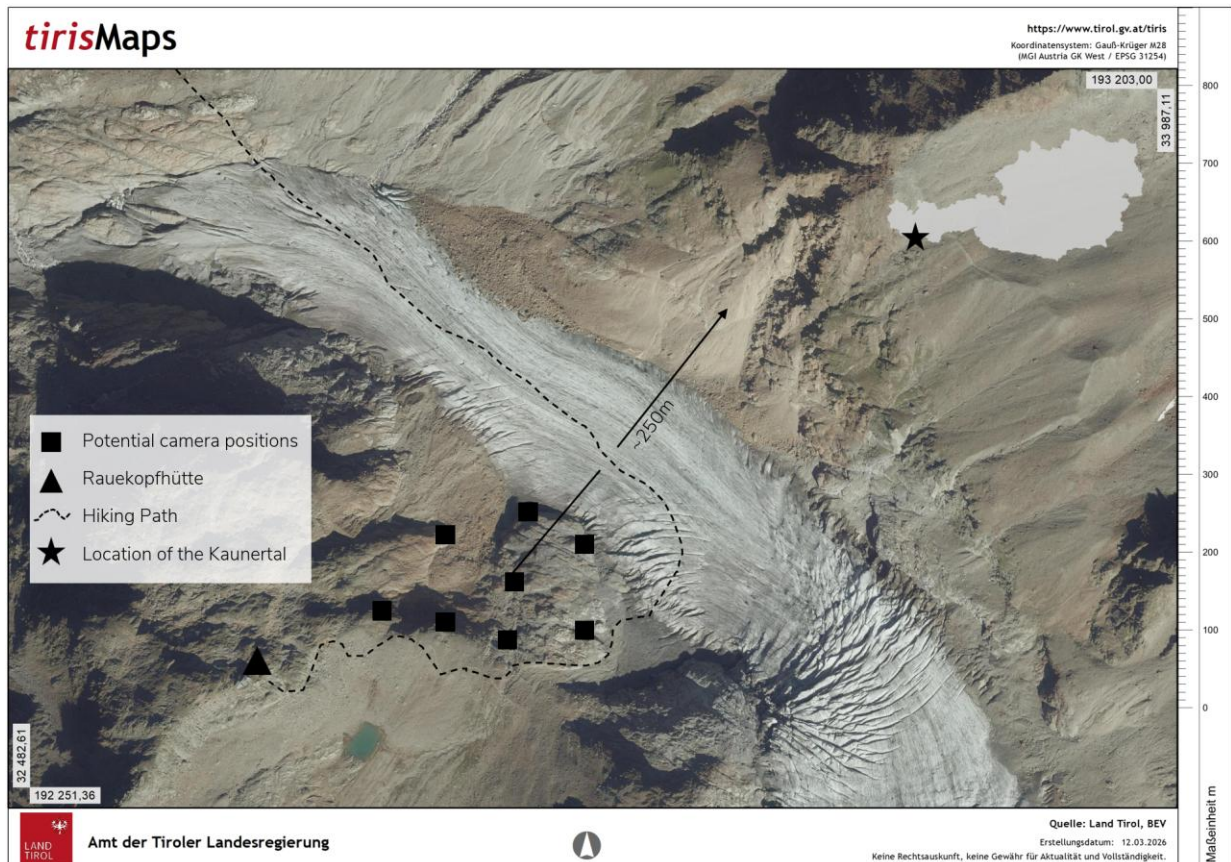


Figure 1. Overview of the study site and location of the Kaunertal in Austria (map adapted from tirol.gv.at)

The monitored rockwall extends approximately **300 m horizontally** and **about 300 m vertically**, covering a **total area of roughly 45,000 m²** between **elevations of 2400 and 2800 m.a.s.l.** (Fig. 1 and 2). Previous LiDAR-based investigations **quantified geomorphic work of approximately 159 GJ per year on this single rock slope**, with the majority generated by rockfall processes (Vehling et al. 2017). However, these observations **provide only cumulative estimates of surface change** and do **not resolve the timing, frequency, or triggering mechanisms** of individual rockfall events.

The rockfall deposition area is located in **close proximity to the popular hiking route leading to the Rauhekopfhütte** (DAV – Frankfurt) mountain hut, emphasising the relevance for local hazard assessment. In addition, Vehling et al. (2017) identified a **large unstable rock mass (~1.25 million m³) on the slope**.

The site is particularly suitable for photogrammetric monitoring because an opposing valley flank located approximately 200 m from the rockwall provides **ideal positions for camera installation**. The research team has **more than ten years of field experience** in the area and maintains **established collaborations with local stakeholders** including the Naturpark Kaunergrat, the municipality of Kaunertal, local landowners from Birgalpe, mountain guides, and tourism organisations.

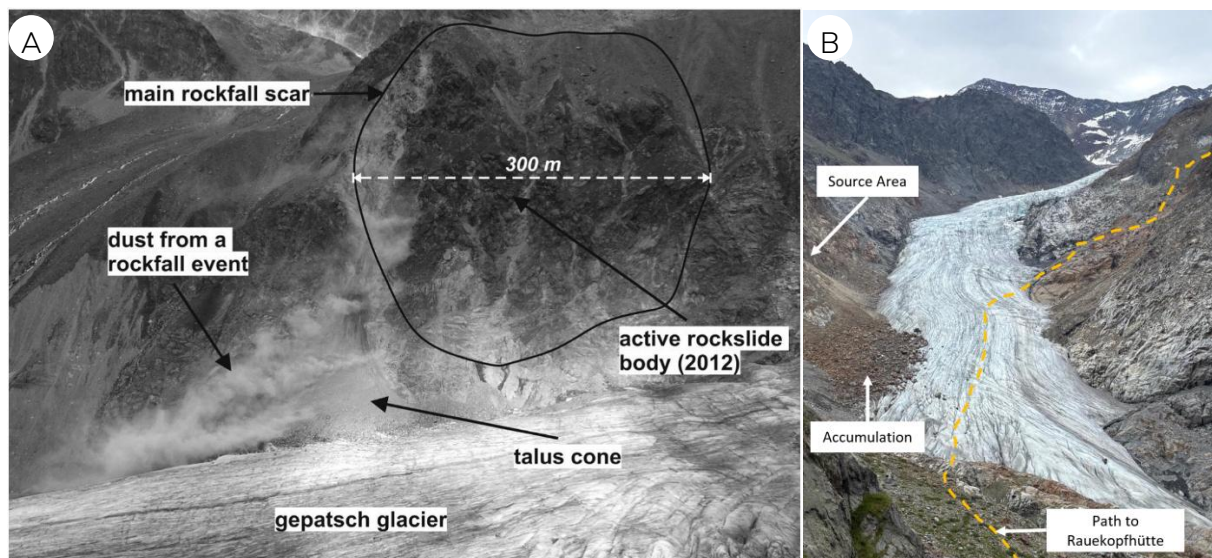


Figure 2. Active rock fall process at Schwarze Wand' in September 2012 (A) recorded by D. Morche, (Vehling et al. 2017) and current location of the hiking path to Rauhekopfhütte (B, adapted from rauhekopfhütte.at)

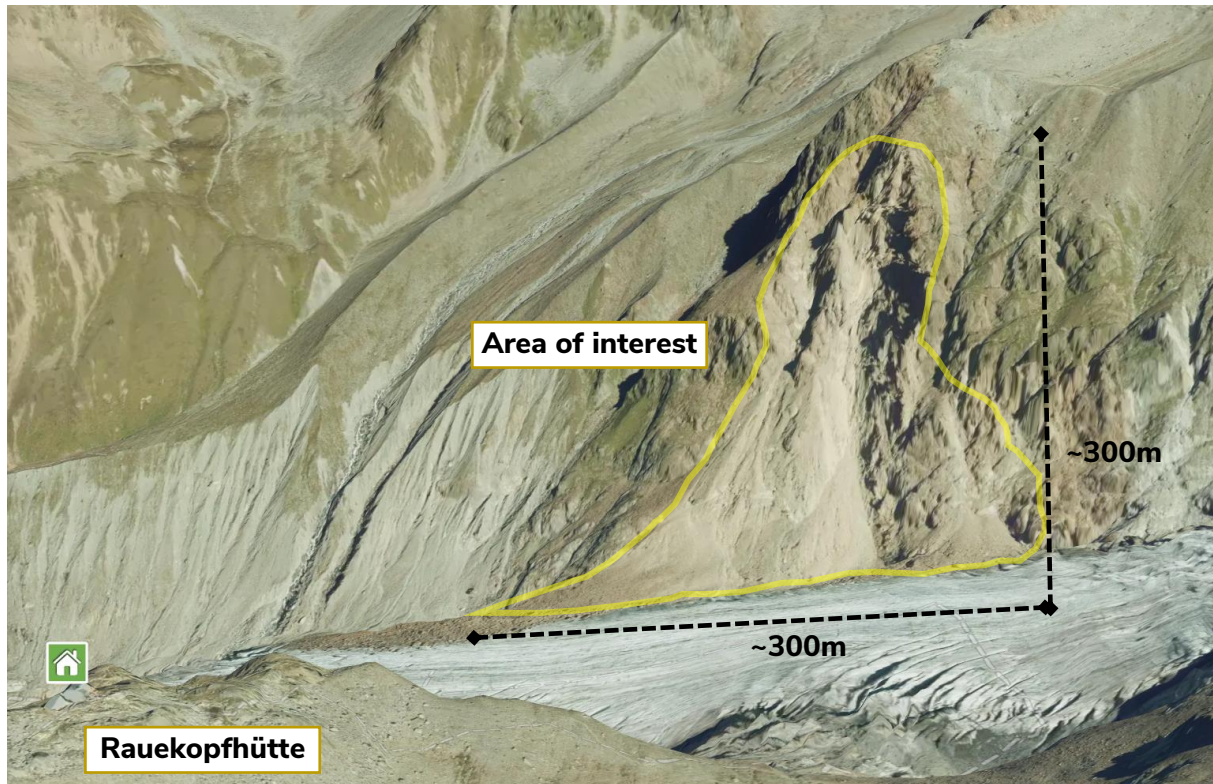


Figure 3. Approximate area of interested – delineated from realitymaps.com

4. Monitoring concept and methodology

The project will deploy a **network of autonomous time lapse cameras** designed for long term operation in high mountain environments. **Seven high resolution cameras equipped with full frame sensors and 135 mm lenses** will be installed at stable vantage points opposite the monitored rockwall. An **additional overview camera** will document the entire slope and provide contextual information for change detection. The camera configuration ensures **high image overlap** and stable multi-view geometry required for reliable photogrammetric reconstruction.

Each camera unit will be housed in a weatherproof enclosure and powered by a solar energy system consisting of a 150 W solar panel, charge controller, and lithium battery. Image acquisition will be controlled by a Raspberry Pi based system with real time clock modules. **Cameras will capture images every hour during daylight hours between approximately 07:00 and 19:00 from June to November.**

The monitoring geometry and camera configuration are expected to achieve a **ground sampling distance of approximately 2–5 cm**. Photogrammetric processing will produce dense point clouds of the rockwall surface, from which **digital elevation models with grid resolutions of approximately 5–10 cm** will be derived. This spatial resolution enables the detection of small rockfall events while also capturing larger slope failures.

The photogrammetric workflow will include the following steps:

- **Acquisition of time lapse image sequences** from the camera network

- **Photogrammetric reconstruction**
- **Generation of dense point clouds** and **digital elevation models**
- Detection of **surface changes through DEM of Difference analysis**
- **Identification and quantification of rockfall events**
- Compilation of a **rockfall event database including timing, location, and volume**
- **Statistical analysis of rockfall occurrence** in relation to environmental drivers

Additional **baseline datasets will be collected through UAV based photogrammetry twice per season** and a **terrestrial laser scanning survey during installation**. Meteorological data from nearby weather stations operated by regional partners will be used to analyse potential triggering factors.

5. Expected outcomes and relevance

The project will produce a unique high resolution dataset documenting rockwall dynamics in a rapidly deglaciating alpine environment. Key scientific outputs will include:

- a detailed **rockfall event catalogue with sub daily temporal resolution**
- a **four dimensional dataset of rockwall evolution** derived from repeated photogrammetric models
- **quantitative estimates of rockfall volumes and rockwall retreat rates**
- **improved understanding of environmental controls** on rockfall activity in glacier forelands

The project is expected to result in **two peer reviewed publications** addressing paraglacial rockwall dynamics, methodological advances in photogrammetric monitoring, and environmental drivers of rockfall activity.

Beyond its scientific contribution, the project has direct **relevance for natural hazard assessment**. The monitored rockwall is located above a frequently used hiking route, and **improved knowledge of rockfall dynamics can inform risk management and communication with local stakeholders**. The monitoring concept developed in HIROCK is designed to be cost efficient and scalable, making it transferable to other glacier forelands, permafrost rockwalls, and mountain infrastructure corridors.

6. Work plan

The project will be implemented through five main work packages.

- **WP1 Site preparation and camera installation**
Installation of camera network, baseline TLS survey, and system calibration.
- **WP2 Time lapse monitoring and photogrammetric processing**
Continuous image acquisition and generation of high resolution photogrammetric models.
- **WP3 Rockfall detection and quantification**
Identification of rockfall events using DEM of Difference analysis and compilation of the rockfall inventory.

- **WP4 Environmental driver analysis**
Statistical analysis of rockfall activity in relation to meteorological conditions and glacier retreat.
- **WP5 Stakeholder engagement and knowledge transfer**
Communication of results to regional stakeholders, integration into science outreach activities, and dissemination of open datasets.

7. Project team

The project will be led by **Dr Stefan Haselberger (University of Graz)**, who has extensive experience in high mountain geomorphology and glacier foreland research. The project will benefit from collaborations with:

- **Prof Martin Mergili (University of Graz)**, expert in modelling mass movements in alpine environments
- **Prof Anette Eltner (TU Dresden)**, specialist in photogrammetry and geomorphic monitoring
- **Robert Krüger (TU Dresden)**, expert in field deployment of photogrammetric monitoring systems

Additional collaboration with regional stakeholders including Naturpark Kaunergrat, the municipality of Kaunertal, and alpine clubs will ensure strong links between scientific research and local knowledge.

The **proposed pilot phase in 2026** will establish the monitoring infrastructure and generate the first high resolution datasets. These results will form the **basis for a larger follow up project** including a doctoral research position focusing on long term paraglacial rockwall dynamics.

8. References

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